Deep-and-Narrow MoE + More Heads + Partial RoPE

Trade width for depth, add query heads, rotate half

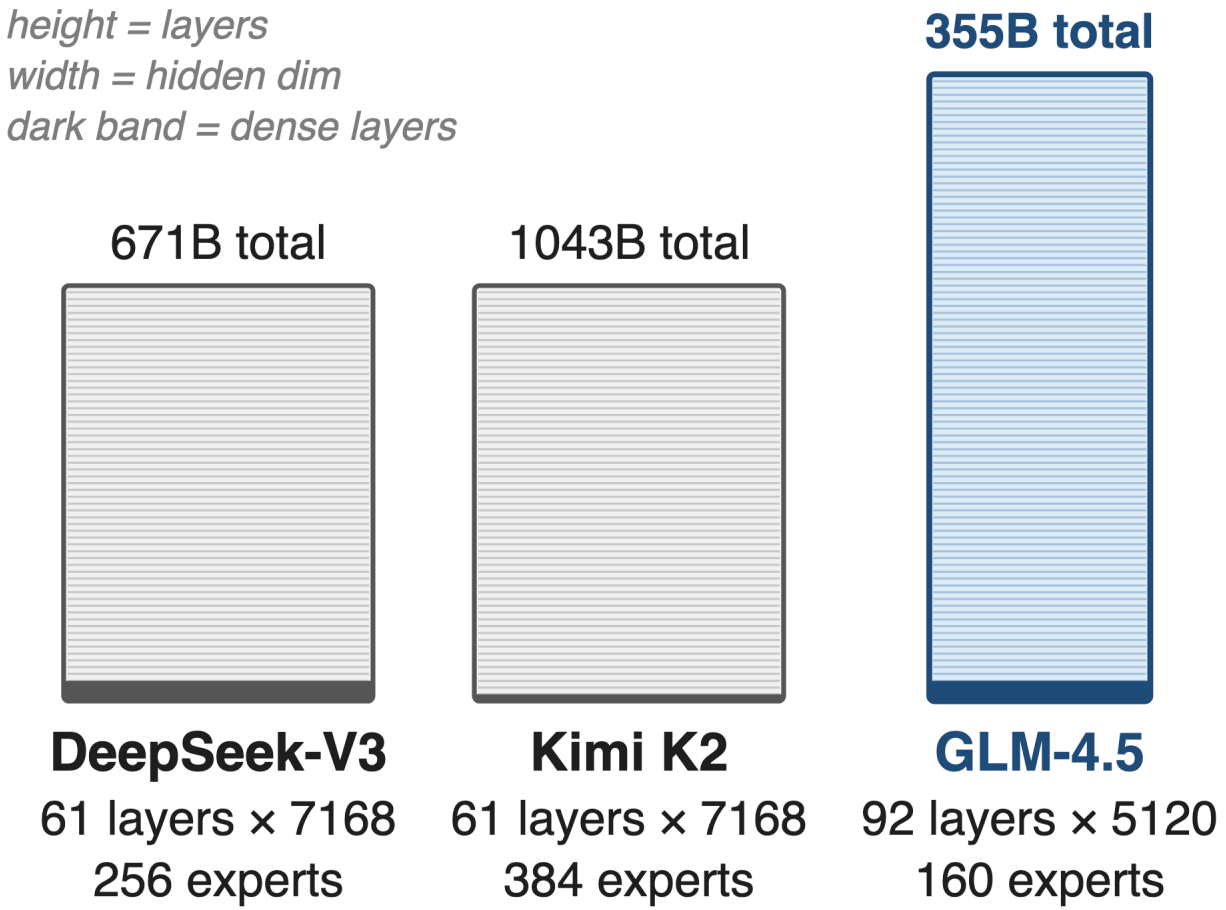
Zhipu AI & Tsinghua · GLM-4.5

Explainer deck · Industrial Pretraining Architecture & Infra Tech Landscape (2026)

Model Shape & Depth Scaling · arxiv.org/abs/2508.06471

Three ~32B-active MoEs, three different shapes

- Similar budget: **32–37B** active per token
- DeepSeek-V3, Kimi K2: **61** layers × **7168**
- GLM-4.5: **92** layers × **5120**, only **160** experts



Width costs weights: K2 holds 1,043B

1,043B

total parameters behind Kimi K2's 32B active: 2.9× GLM-4.5's 355B

GLM-4.5 tech report, arXiv 2508.06471, Table 1 (Kimi K2's own Table 2: 1.04T total, 32.6B active)

Loss-only search rewards width, punishes heads

- **More experts** (K2 sparsity law)
- Same loss at **1.69×** fewer FLOPs
- **256 → 384 experts, 1.04T weights**

- **More heads** (K2 ablation)
- Only **0.5–1.2%** lower validation loss
- **+83% inference FLOPs** at 128K: cut to **64**

Both choices were judged by validation loss alone

Insight: depth and heads buy reasoning, not loss

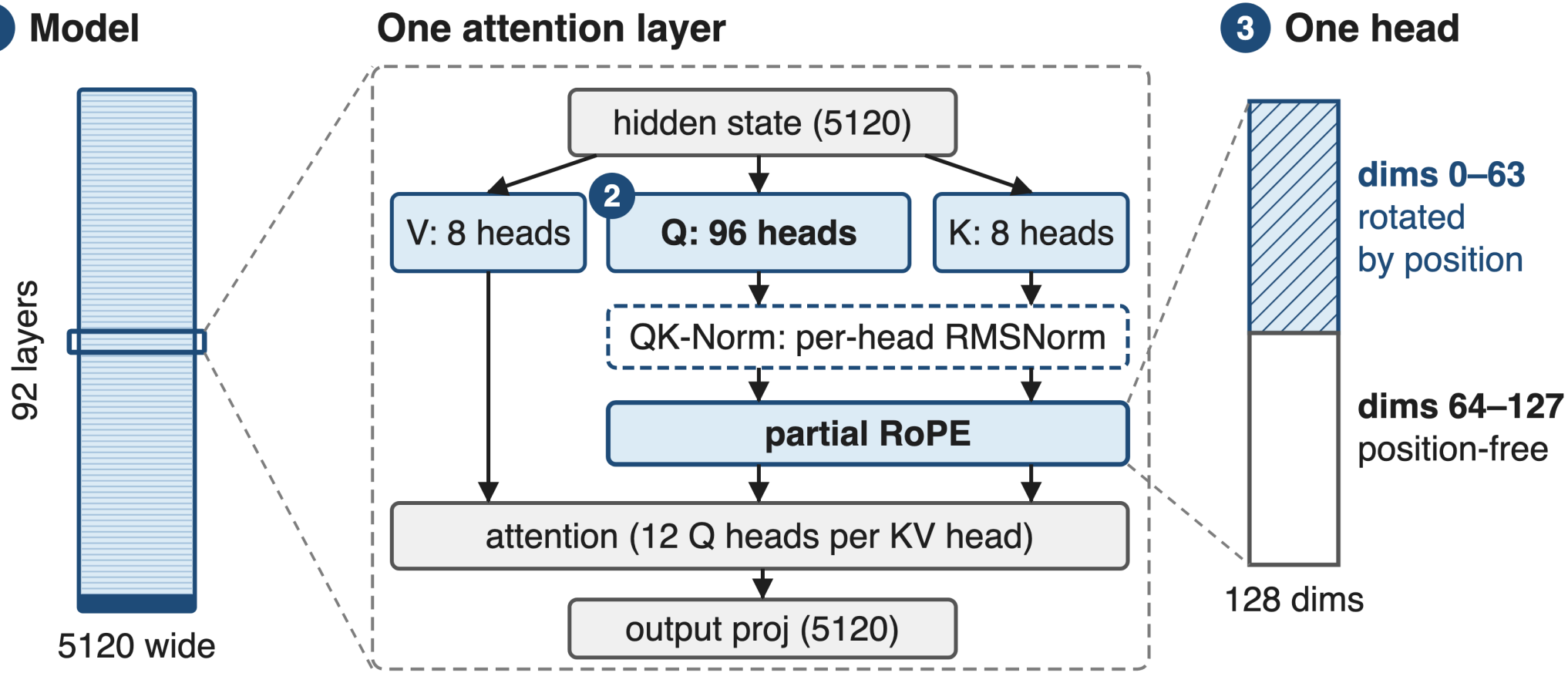
- Deeper models showed **better reasoning**
- **2.5×** more heads: same training loss, higher **MMLU / BBH**

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Loss is the wrong yardstick: spend the active budget on depth and heads

GLM-4.5 reshapes the model at three levels



Block order as in transformers' modeling_glm4_moe.py: projections, QK-Norm, partial RoPE, GQA attention

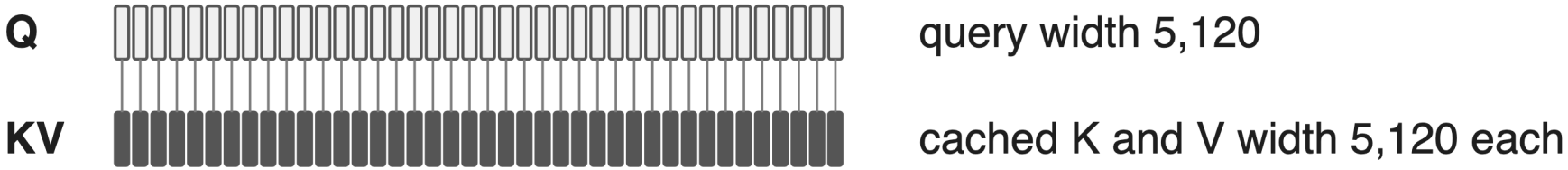
❶ Same active budget, 31 more narrower layers

	DeepSeek-V3	Kimi K2	GLM-4.5
❶ Layers (dense + MoE)	61 (3 + 58)	61 (1 + 60)	**92 (3 + 89)**
Hidden dim	7168	7168	**5120**
Routed experts (active)	256 (8)	384 (8)	**160 (8)**
❷ Attention type	MLA	MLA	**GQA**
Query / KV heads	128 / 128	64 / 64	**96 / 8**
❸ RoPE dims per QK head	64 of 192	64 of 192	**64 of 128**
QK-Norm (deck 66)	no	no	**yes**
Total / active params	671B / 37B	1043B / 32B	**355B / 32B**

GLM-4.5 Table 1; DeepSeek-V3 §4.2; Kimi K2 Table 2; RoPE dims from the public HF configs

❷ 96 query heads share just 8 KV heads

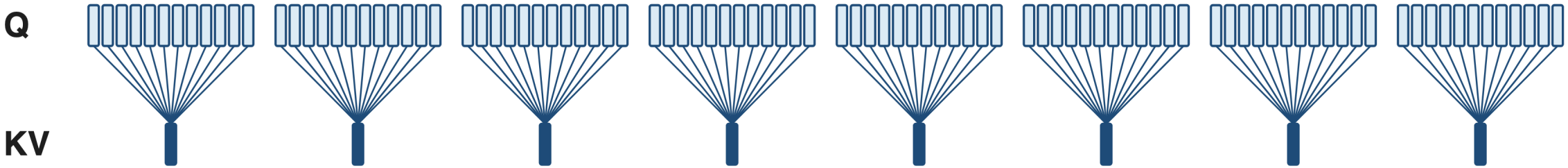
(a) Width-matched layout: 5120 / 128 = 40 heads



query width 5,120

cached K and V width 5,120 each

(b) GLM-4.5: 96 query heads, 8 KV heads



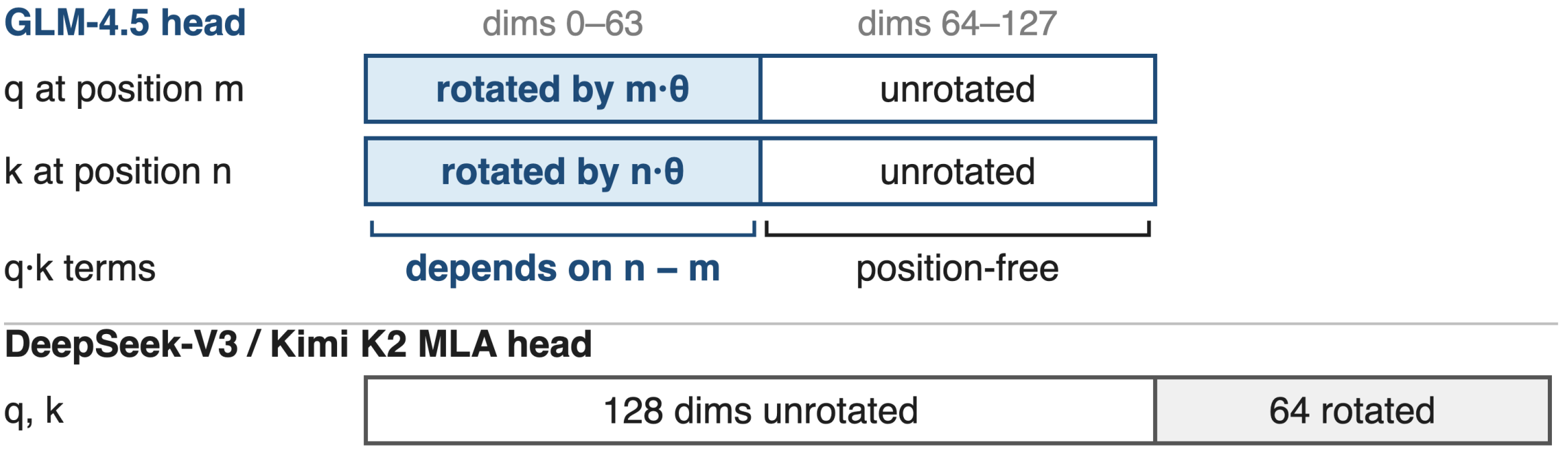
query width 96 × 128 = 12,288

cached K and V width 8 × 128 = 1,024 each

12 query heads per KV head

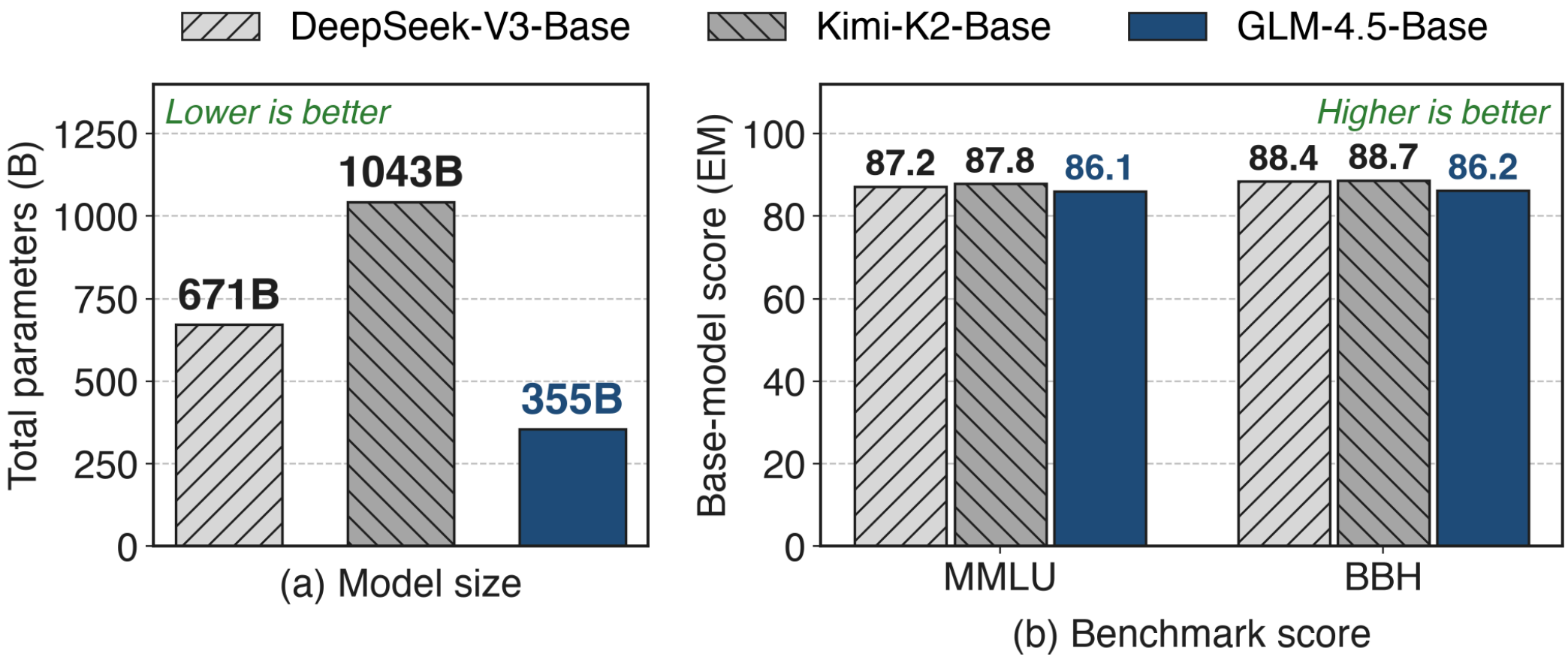
2.4× the width-matched head count (paper: "2.5×"); cached KV stays at 8 heads

❸ Partial RoPE rotates only half of each head



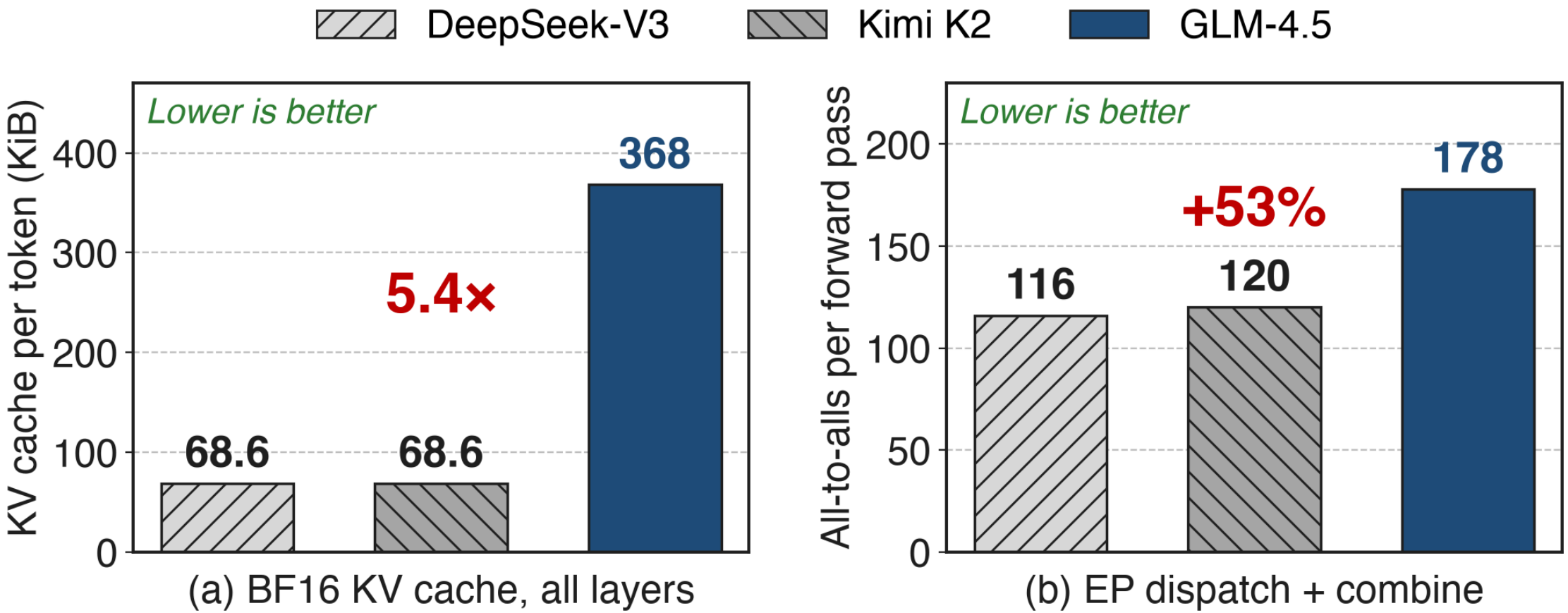
Each head keeps a 64-dim channel that matches content at any distance

A third of K2's weights, within 2.5 points



GLM-4.5 Table 2 (GLM's internal eval). Not an ablation: 23T vs 14.8T vs 15.5T training tokens

The bill: 5.4× KV cache, 53% more all-to-alls



Analytic, from published configs (BF16 KV cache; 2 all-to-alls per MoE layer). Dispatched bytes per token: only +10%

GLM-5 cut depth to 78 and moved to MLA

- **GLM-5**: 744B total, 40B active
- **78** layers, to cut EP communication
- MLA for memory; **64** heads, **6144** wide

- **Open questions**
- No published ablation numbers
- Shape confounded with data and Muon

Stated reasons: fewer layers for EP traffic, MLA for memory

Summary

- Wide template: K2 holds **1,043B** weights for **32B** active
- **Insight**: depth and heads buy reasoning that loss misses
- GLM-4.5: **92** × **5120**, **96** Q / **8** KV heads, half-RoPE
- **355B** total; base MMLU/BBH within **2.5** points of K2

https://arxiv.org/abs/2508.06471